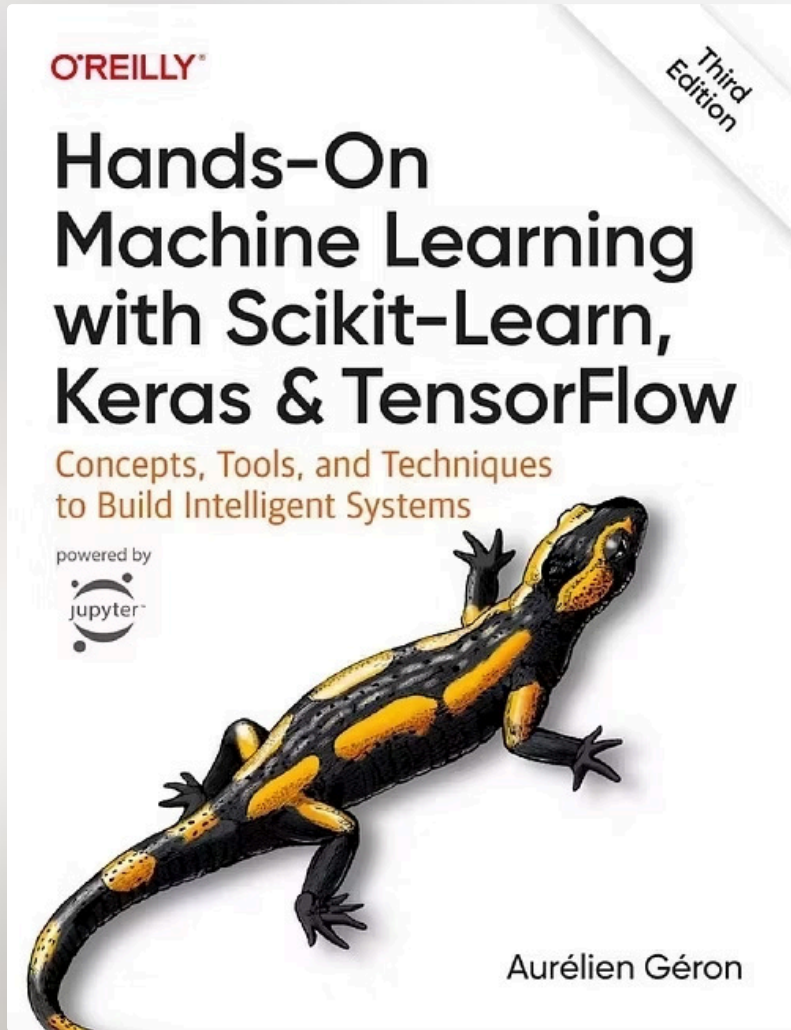




Machine Learning

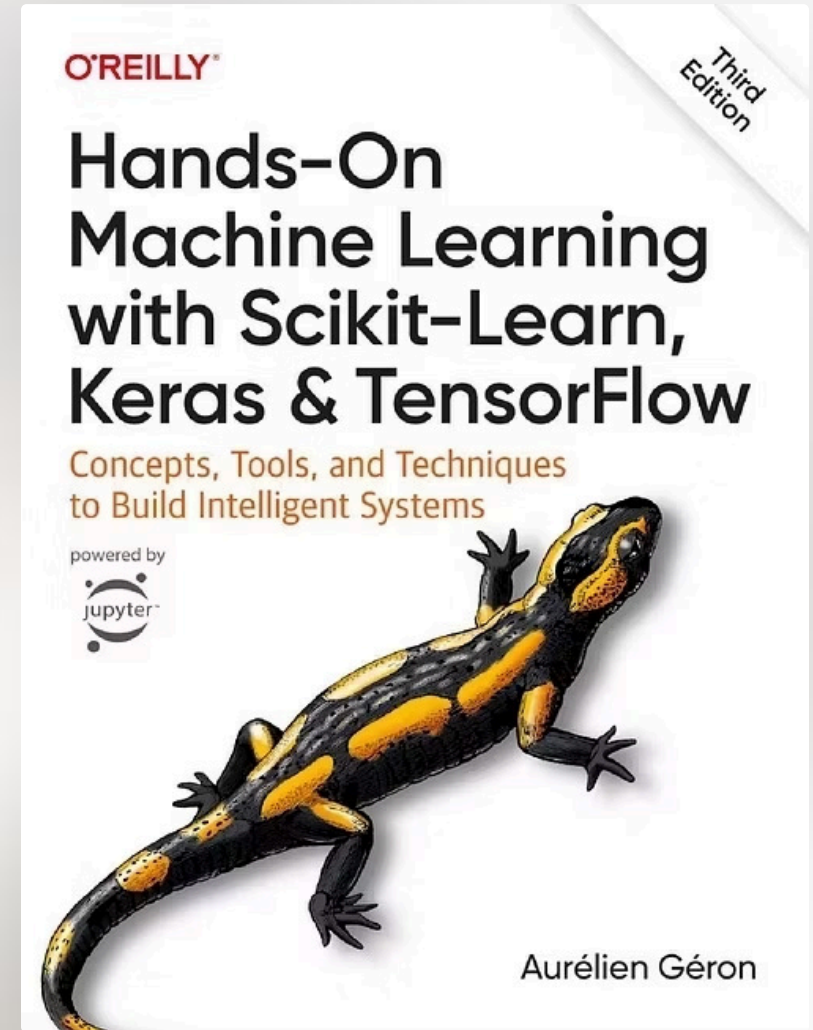
Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

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Machine Learning

Chapter 1 – The Machine Learning Landscape





Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. **Testing and Validating**

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back: Reviewing the Chapter So Far
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Testing and Validating

Hyperparameter Tuning and Model Selection, Data Mismatch

Testing and Validating Machine Learning Models

Goal:

The only way to assess how well a model will **generalize to new, unseen data** is to **test** it on new cases.



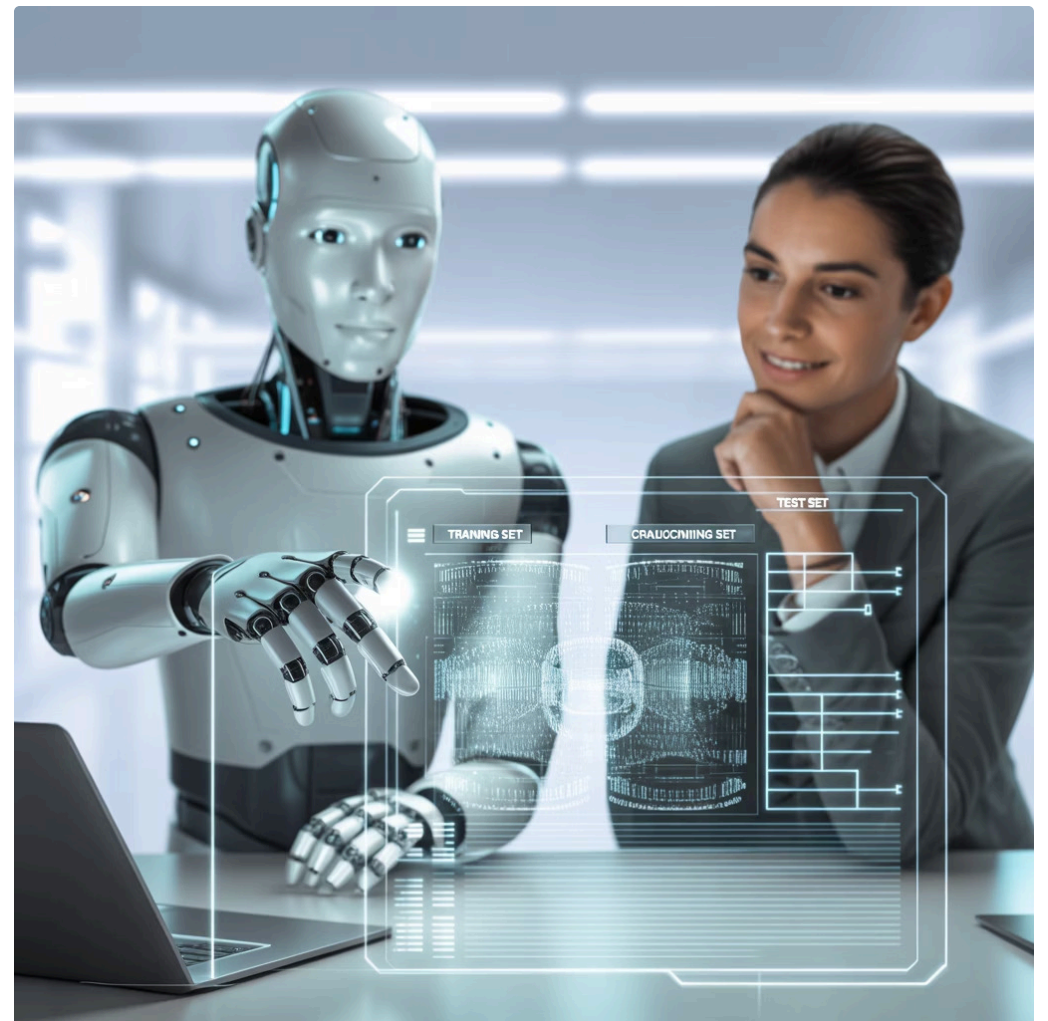
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Production Testing vs. Evaluation Testing



Production Testing:

Testing the model in production works, but if the **model performs poorly**, user complaints will occur, making this option **risky**.



Better Approach:

Split the data into two sets:

- **Training Set:** Used to train the model.
- **Test Set:** Used to evaluate the model's performance.

Generalization Error

Definition:

The **generalization error** (or **out-of-sample error**) is the **error rate** when the **model** is **tested** on **data** it has **never seen before**.

Why It Matters:

Evaluating your **model** on the **test set** gives an **estimate** of its **ability** to **generalize** to **new data**.



Underfitting vs. Overfitting

1

Low Training Error, High Generalization Error:

If the model performs well on the **training set** (low training error) but poorly on the **test set** (high generalization error), the model is **overfitting** the training data.

2

Overfitting Warning:

The model has **learned the training data too well**, capturing **noise and irrelevant details** rather than general patterns.



Test Set Split: Common Practice



80%

Training Data

Common split for
model training

20%

Test Data

Held out for
evaluation

Dataset Size Considerations:

For **large datasets** (e.g., 10 million instances), even holding out 1% for testing provides a sufficiently large test set.

Key Takeaways

1

Testing and Validation:

Split your data into a **training set** and a **test set** to evaluate model performance.

2

Generalization Error:

Use the **test set** to **estimate the model's ability to generalize to new data**, and watch out for **overfitting** if **training error is low** but **generalization error is high**.

3

Data Split:

Typically, **80%** for training and **20%** for testing, though this depends on **dataset size**.



Next: Hyperparameter Tuning and Model Selection

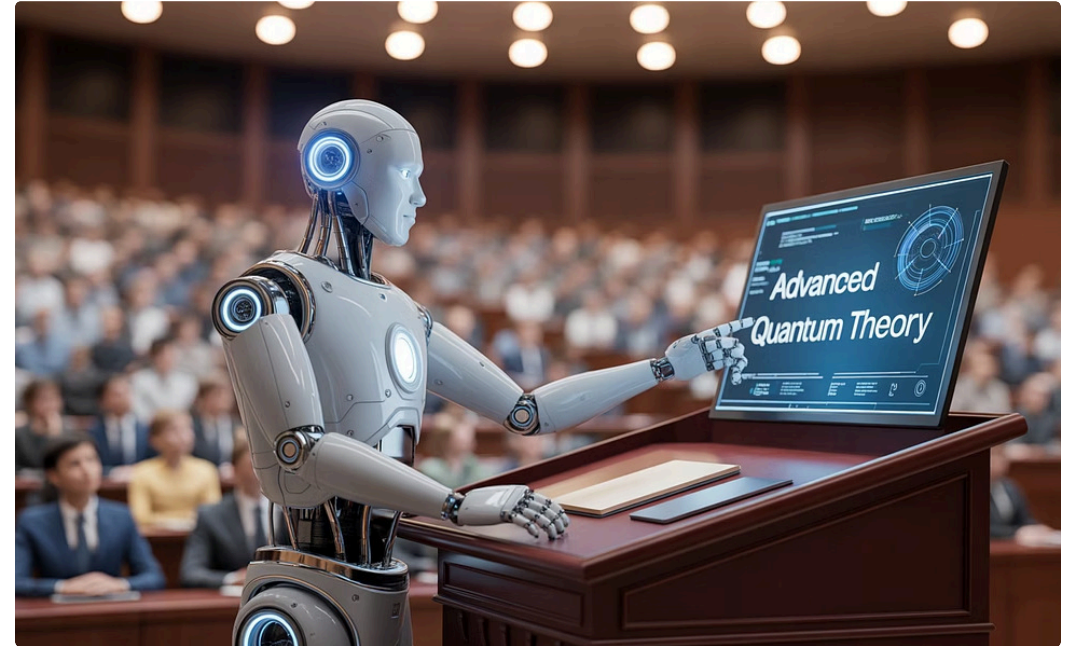


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts